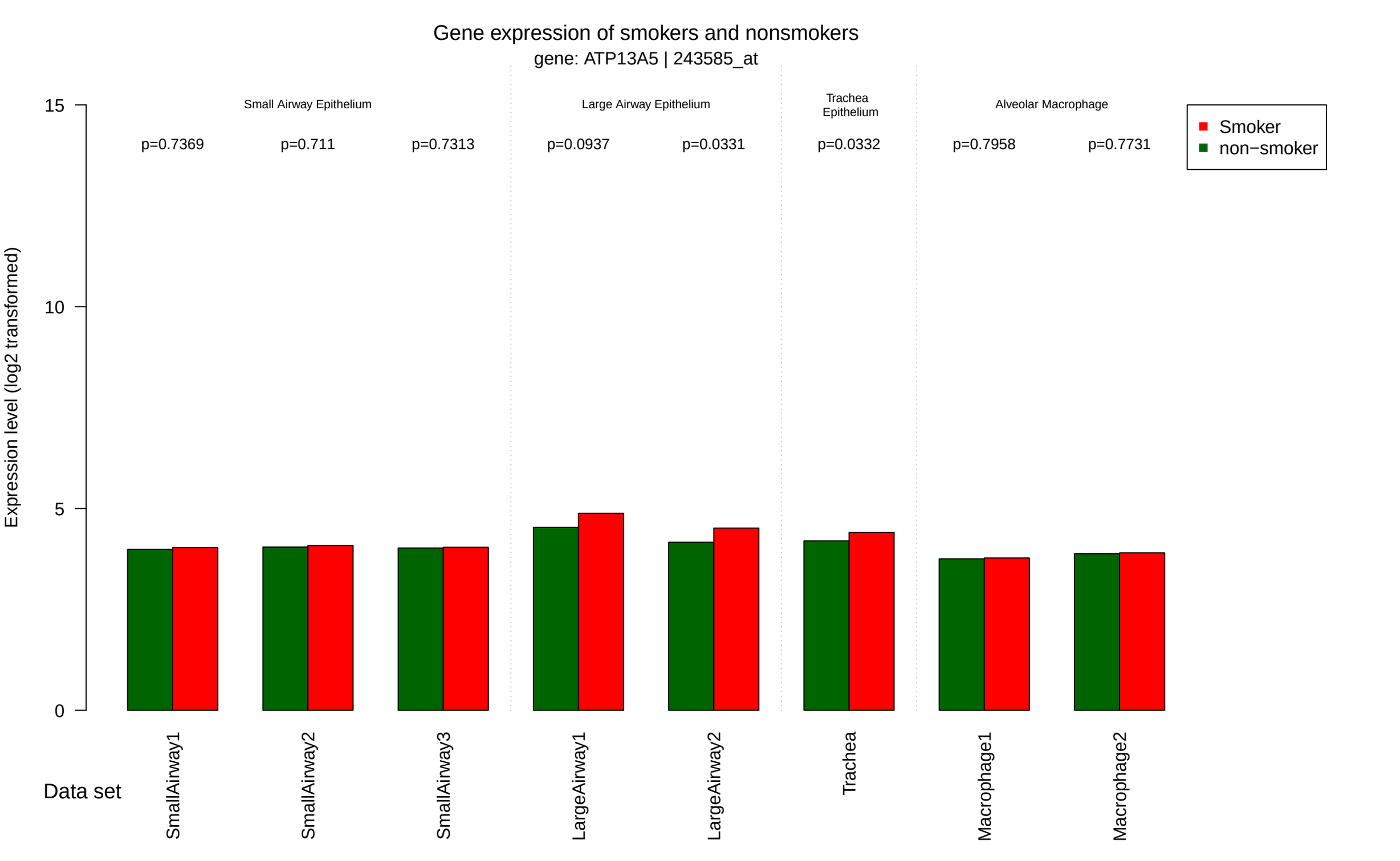
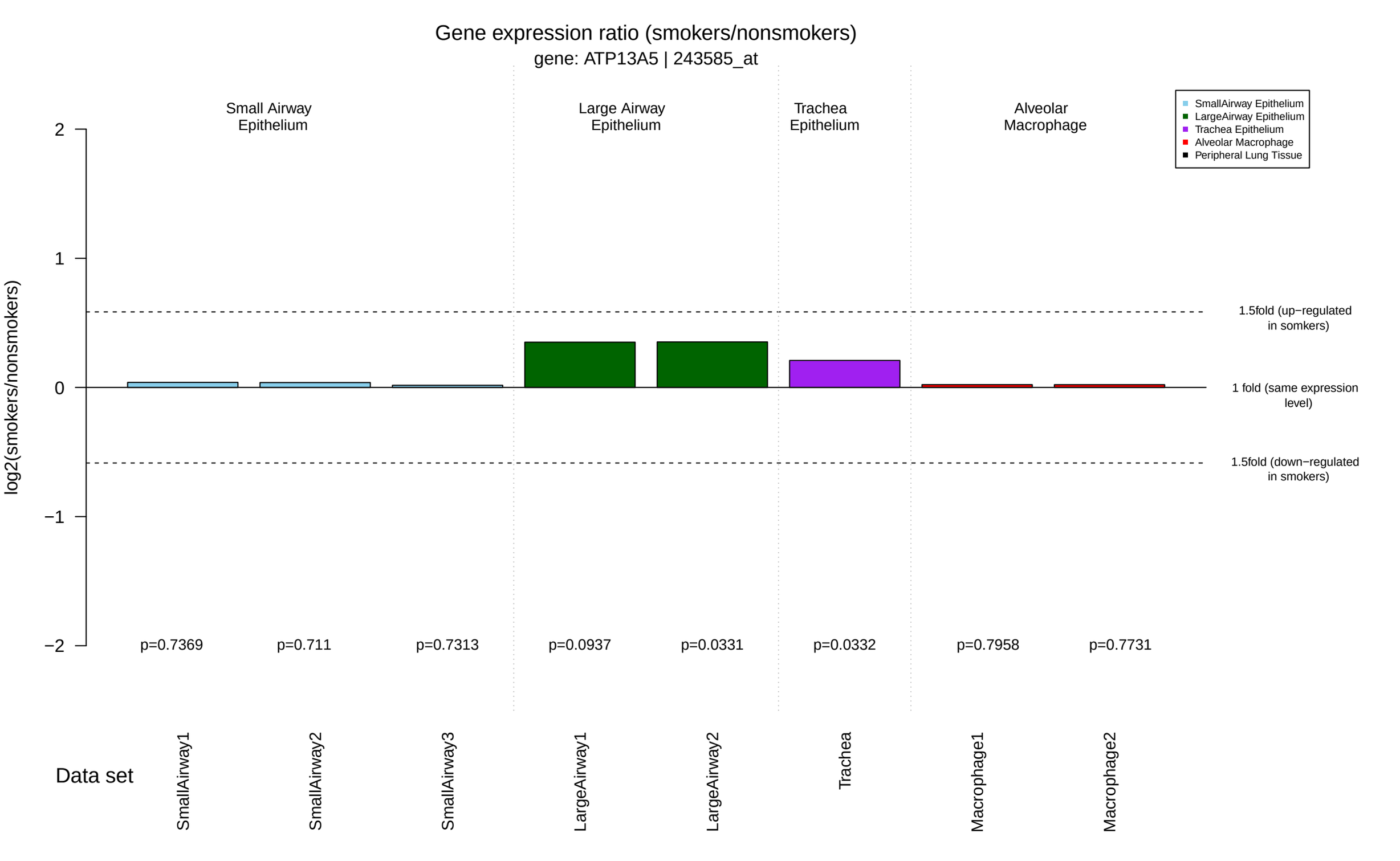
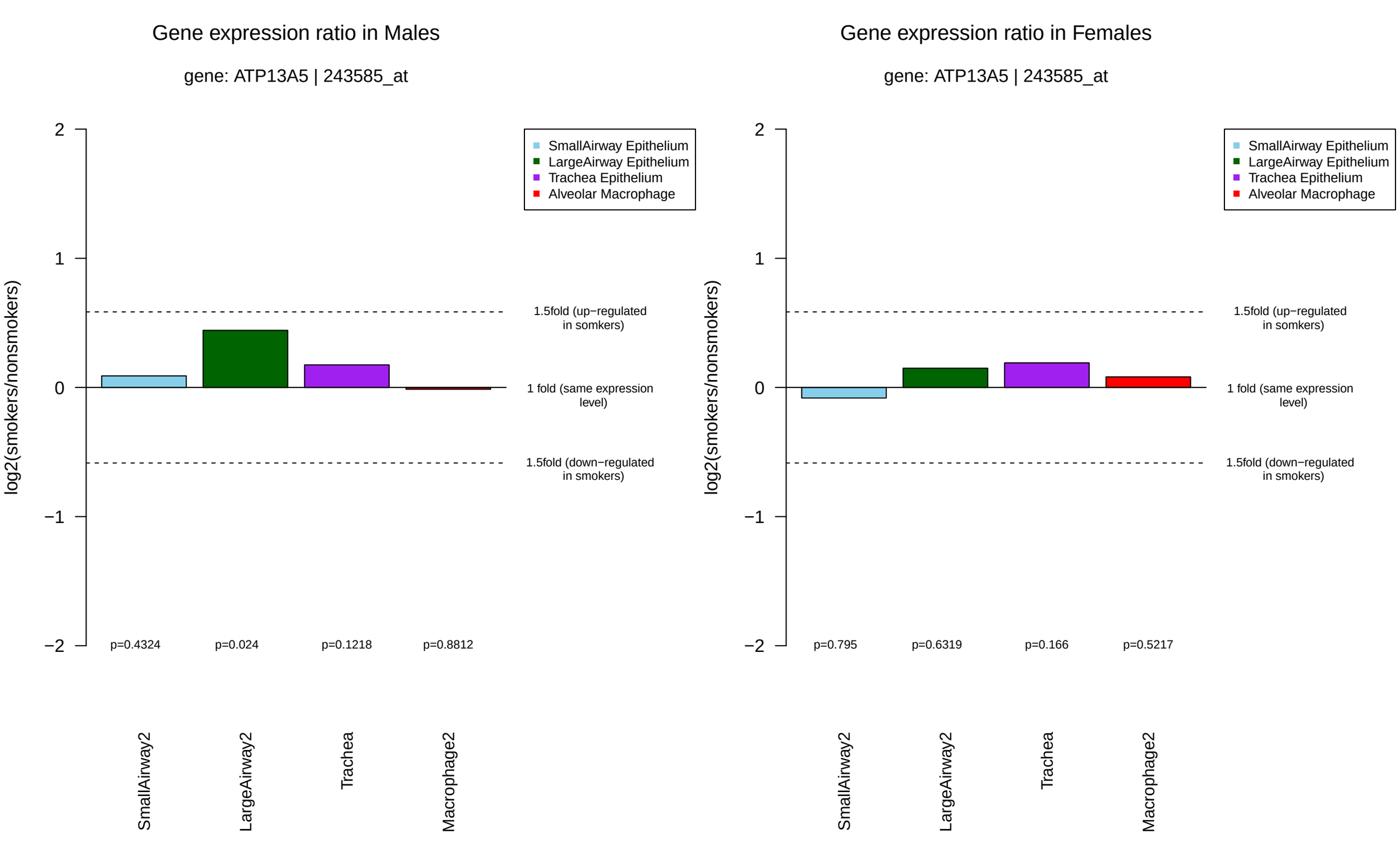
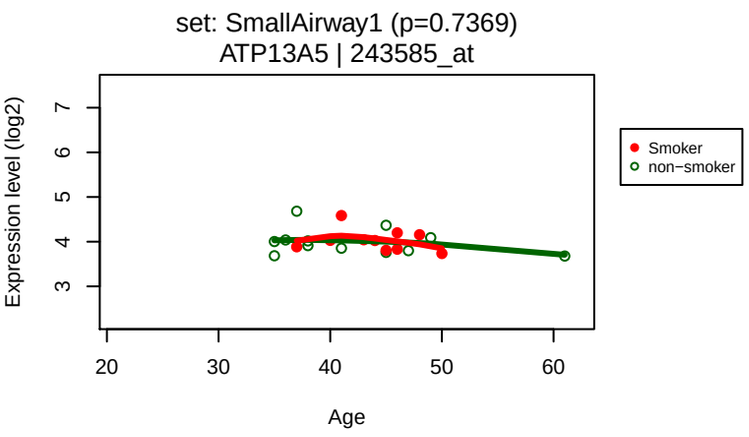
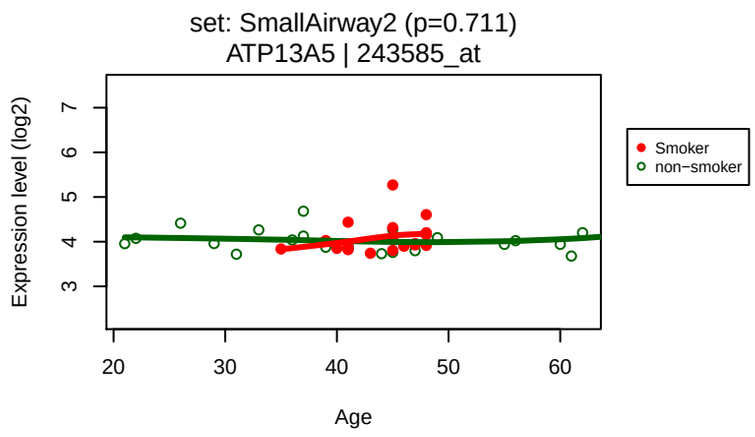
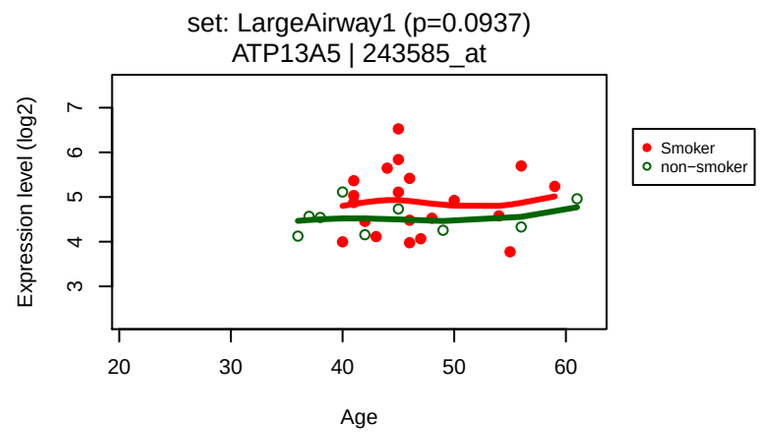

Gene expression of smokers and nonsmokers
gene: ATP13A5 | 243585_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7369
p=0.711
p=0.7313
p=0.0937
p=0.0331
p=0.0332
p=0.7958
p=0.7731
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

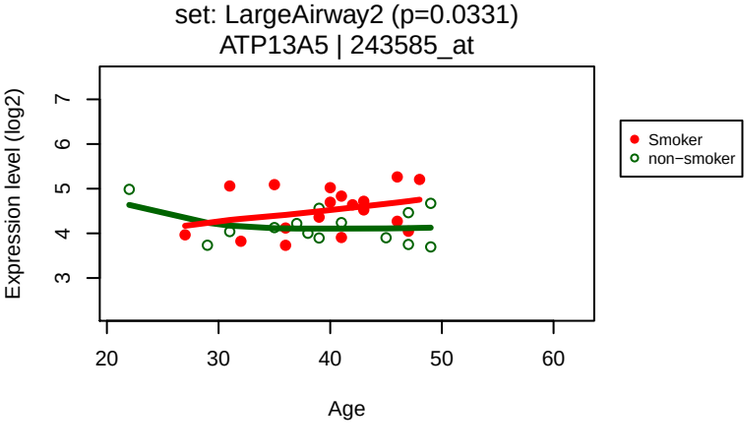

Gene expression ratio (smokers/nonsmokers)
gene: ATP13A5 | 243585_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7369
p=0.711
p=0.7313
p=0.0937
p=0.0331
p=0.0332
p=0.7958
p=0.7731
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

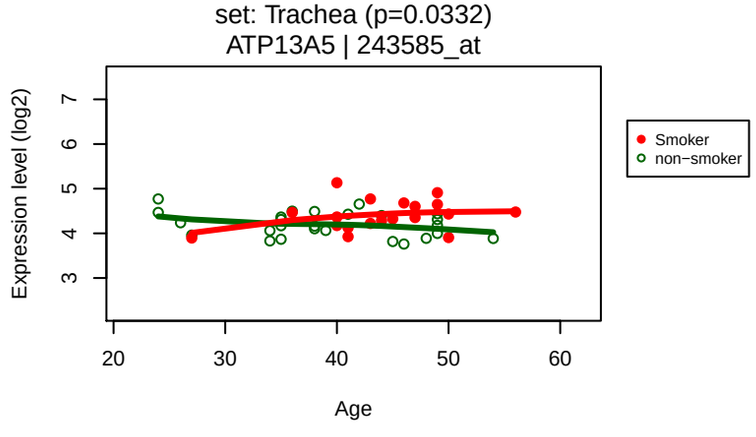

Gene expression ratio in Males
gene: ATP13A5 | 243585_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4324
p=0.024
p=0.1218
p=0.8812
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP13A5 | 243585_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.795
p=0.6319
p=0.166
p=0.5217
SmallAirway2
LargeAirway2
Trachea
Macrophage2

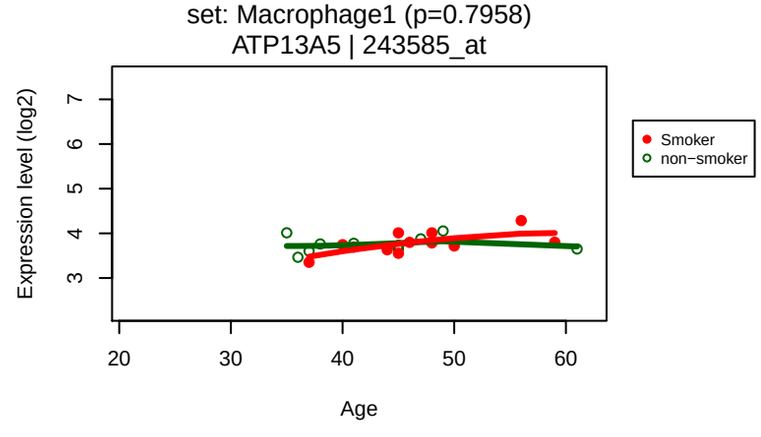

set: SmallAirway1 (p=0.7369)
ATP13A5 | 243585_at
Expression level (log2)
Age
Smoker
non−smoker

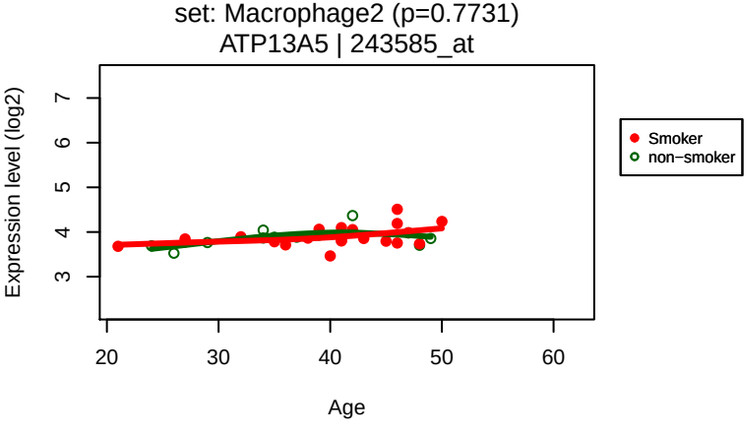

set: SmallAirway2 (p=0.711)
ATP13A5 | 243585_at
Expression level (log2)
Age
Smoker
non−smoker

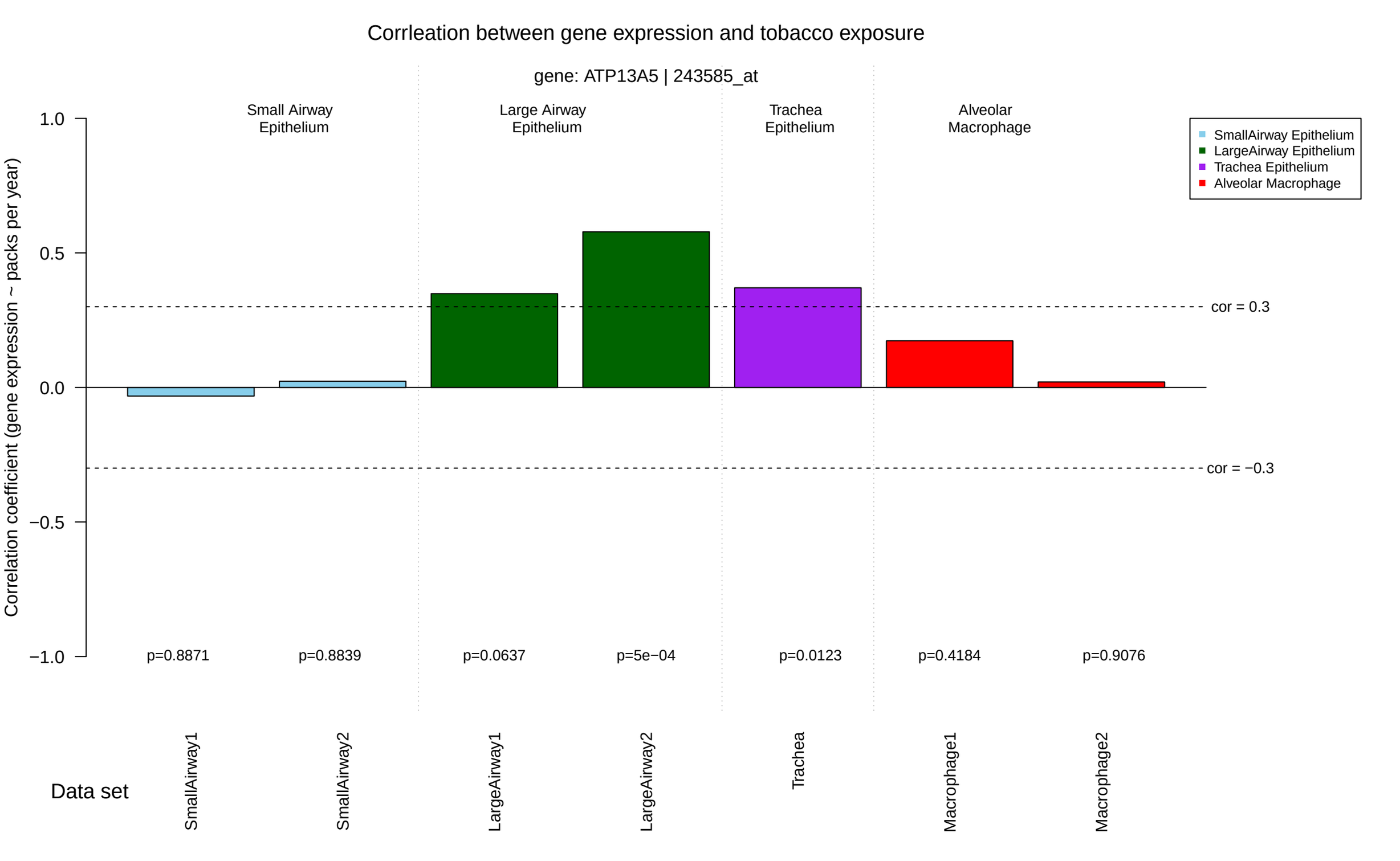

set: LargeAirway1 (p=0.0937)
ATP13A5 | 243585_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0331)
ATP13A5 | 243585_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0332)
ATP13A5 | 243585_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7958)
ATP13A5 | 243585_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7731)
ATP13A5 | 243585_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP13A5 | 243585_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8871
p=0.8839
p=0.0637
p=5e−04
p=0.0123
p=0.4184
p=0.9076
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2